# Ridge Regression Using Gradient Descent.

## Vector Form Loss

$$L = \sum_{i=i}^{n} (y_i - \hat{y}_i)^2$$

$$= (XW - Y)^T (XW - Y) + \lambda ||W||^2$$

$$= (XW - Y)^T (XW - Y) + \lambda W^T W$$

$$W_0 = W_0 - \eta \frac{\partial L}{\partial W_0}$$

$$W_1 = W_1 - \eta \frac{\partial L}{\partial W_1} \quad \ldots \quad W_n = W_n - \eta \frac{\partial L}{\partial W_n}$$

$$L = \frac{1}{2} (XW - Y)^T (XW - Y) + \frac{1}{2} \lambda W^T W$$

$$= \frac{1}{2}(W^T X^T - y^T)(XW - y) + \frac{1}{2}\lambda w^T W$$

$$= \frac{1}{2}\left[W^T X^T X W - W^T X^T y - y^T W X + y^T y\right] + \frac{1}{2}\lambda W^T W$$

$$= \frac{1}{2}\left[W^T X^T X W - 2 W^T X^T y + y^T y\right] + \frac{1}{2}\lambda W^T W$$

$$\frac{\partial L}{\partial W} = \frac{1}{\cancel{2}}\left[\cancel{2} X^T X W - \cancel{2} X^T y\right] + \frac{1}{\cancel{2}} \cancel{2}\lambda W$$

$$= X^T X W - X^T y + \lambda W$$

$$\boxed{\frac{\partial L}{\partial W} = X^T X W - X^T y + \lambda W} = \frac{\Delta L}{\Delta W}$$